Title Page:

A Library of Human Electrocorticographic Data and Analyses

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Conflict of Interest: The author has no competing interests.

Acknowledgements: I am grateful to Jeffrey Ojemann, Marcel den Nijs, Rajesh Rao, and Gerwin Schalk for many years of selfless mentorship during the development of this library. Dora Hermes, Robert Eisinger and Brian Wandell provided many helpful discussions about open data and helped with critical readings of this manuscript. The scientific value of the time spent by our epilepsy patients at Harborview Hospital in Seattle is immense, and I am thankful for their enthusiastic participation. I am financially supported by the Van Wagenen Foundation. Data collection was supported by NSF BCS-0642848 and NIH RO1NS065186. The funders had no role in study design, data collection and analysis, decision to publish or preparation of the manuscript.

Abstract

Electrophysiological data from implanted electrodes in the human brain are rare, and therefore scientific access to it has remained somewhat exclusive. Here we present a freely-available curated library of implanted electrocorticographic (ECoG) data and analyses for 16 benchmark behavioral experiments, with 204 individual datasets from 34 patients made with the same amplifiers (at the same sampling rate and filter settings). In every case, electrode positions have been carefully registered to brain anatomy. A large set of fully-commented analysis scripts to interpret these data using modern techniques is embedded in the library alongside the data. All data, anatomic correlations, and analysis files (MATLAB code) are in a common, intuitive file structure at https://searchworks.stanford.edu/view/zk881ps0522. The library may be used as course material or serve as a starter package for researchers early in their career or for established groups, to modify the analyses and re-apply them in new settings.

Introduction: The manuscript describes data and code libraries, that are the result of an eight-year human electrocorticography (ECoG) research program. This research involved epilepsy patients being monitored for seizure focus localization who volunteered their time in the hospital and participated in a variety of behavioral tasks at the bedside (figure 1). These data provide a unique glimpse into the electrophysiology of the living human brain, measured with subdural electrodes placed in a wide variety of areas.

When compared with other techniques to examine human brain function, ECoG sits in a type of "Goldilocks zone" of experimental scales, with the caveat that electrode placement is stochastic and sparse because of clinical indication. By intrinsically averaging across a population, ECoG is not limited by stochastic responses in the same way that single unit measurements are. Because the spatial scale of cortical columns is not drastically smaller than the exposed ECoG electrode surface, 'local' physiology correlated with a particular brain function is captured. Broadband spectral changes in the ECoG signal have been shown across a variety of brain regions and behavioral tasks to reflect a robust general correlate of neuronal population activity[1], but limited to a very sparse sampling of the cortical surface. Based upon modeling and empirical measurement, our best current understanding is that broadband ECoG changes likely reflect the average input firing rate to the neuronal population beneath each electrode[2,3] (figure 2). This makes ECoG changes robust on a single-event scale (e.g. faster than most perceptions, decisions, and behaviors), so the effect of *immediate context* may be examined on a single trial basis with ~20ms resolution[4,5], rather than *average context* across repeated events.

The experiments performed in this library were tailored to capitalize on occasions where incidental electrode placement coincided with regions known to induce fMRI BOLD changes. By keeping the surgical and experimental parameters fixed over a prolonged period, datasets could be accumulated to cohorts of at least several patients having the same salient brain region coverage for each task. The accompanying MATLAB script library is tailored specifically to identify and extract high-yield features from the ECoG signal (figures 2-4). I believe it will serve as a useful benchmark reference tool for neuroscience research and course teaching.

Results: This library of 16 experiments composed of 204 datasets is available for use without restriction at https://searchworks.stanford.edu/view/zk881ps0522.

Experimental setting: These ECoG data were recorded at the bedside during a monitoring period to localize the seizure focus (where in the brain abnormal activity begins and spreads

from) in epilepsy patients at Harborview Hospital, in Seattle, Washington over a 7-year period (figure 1). Platinum ECoG electrode arrays comprised of grids and strips of electrodes were placed subdurally on the brain surface via craniotomy (anatomic locations were clinically determined by a multi-disciplinary clinical working group, without consideration for research interest). Post-operative radiographic imaging was then correlated with anatomical brain rendering (figure 5). A variety of behavioral tasks were constructed and performed based upon the fMRI literature (Table 1). A subset of tasks that would be expected to engage regions incidentally covered by the clinical ECoG array were selected individually for each patient (e.g. where the ECoG-covered brain regions would be expected to be active, based upon fMRI findings).

Library structure: The library contains a set of zipped folders, with the folder name corresponding to each experiment noted in the left column of table 1. When unzipped, each folder contains a file titled 'README_[experiment name]_dataset_notes'. Each readme file contains: an ethics statement, including IRB information, that must be included in the methodology of any publications; a full reference to the original manuscript(s) where the data were published; the naming conventions for the datafiles, and where they can be found; a description of variables within each datafile; instructions for how to open analysis scripts and initiate analyses and generate figures with analyses plotted on brain surface reconstructions (typically by parsing the commentary of a "master" script).

The main datafiles are in the 'data' subfolder for each experiment. When applicable, the subfolder "ref" contains the reference where the experiment data were originally published. Analyses are initiated from the MATLAB scripts contained in each experimental folder – intermediate signal processing results are saved back into the "data" subfolder, and produced figures are saved into the "figs" subfolder ("figs" is empty until analysis scripts are run). Patient electrode locations are in subfolders named 'locs', and anatomic reconstructions are in a 'brains' subfolder (if renderings were attainable). In some experimental folders, electrode locations and renderings are not separate files and are instead located as variables in the main datafile for each patient in the 'data' subfolder. Anytime spectral decoupling is performed, a subfolder "dc_files" is included within the experiment directory ("fixation_PAC", "fingerflex", "faces_noise", "faces_basic"). When a large number of intermediate files are saved and many figures per patient are generated, the "figs" and "data" subfolders have their own subfolders corresponding to each patient (and named accordingly).

In addition to the experimental folders noted in table 1, there are four additional folders in the root directory of the library that the experimental folders make use of (these four should be downloaded and added to the user's MATLAB path): (1) A folder of general MATLAB scripts, used commonly for analyses is in many different experiments, titled 'toolbox'. (2) A folder titled 'loc' that contains the scripts for estimating electrode location on the standard MNI atlas brain (in Talairach coordinates) from x-ray, and for plotting results from this localization[6] (figure 5a). (3) A folder titled 'ctmr' contains code for estimating electrode positions by fusing pre-operative MRI to post-implant CT[7] (figure 5b). This folder also has many accessory scripts for plotting activity to the patient-specific brain surface rendering (obtained by segmentation of the pre-operative MRI). (4) A folder titled 'xs_tools' contains code for estimating and plotting post-implantation CT electrode positions to pre-operative axial MRI slices[5] (figure 5c).

Basic phenomena observed in the ECoG measurement: Extending the work of Nathan Crone and colleagues[8,9], we found spatially separable motor-associated power changes in of the ECoG voltage measurement[10]. In low frequencies (up to ~30Hz), there are spatially-broad movement-associated decreases in power in oscillatory processes (brain rhythms) over the peri-central cortical surface. Within the region of low frequency power change, there are spatially-focal, somatotopically-discrete, high frequency (above ~50Hz) power increases (analysis scripts in experiment folder "motor_basic" – effect also seen in figure 6a).

Further examination of the high-frequency changes reveals broadband power increases (present in lower frequencies, but obscured by oscillations) that look like a random walk in the voltage timeseries, were shown to obey Gaussian statistics (revealed by a power-of-2 structure in the power spectral density, with $P \sim 1/f^4$), such as might be observed in the aggregate firing of a population of neurons (figure 2a-d,g)[1-4] (experiments 'fixation_highfreq' and 'fixation_pwrlaw'). Broadband increases in this 1/f process are tightly correlated with activation of very local circuits in a wide variety of settings: early visual cortex while viewing visual stimuli[11] (experiment 'visual_search', figure 3); ventral temporal category-selective regions while viewing images of faces and other objects[5,12] (experiments 'faces_basic' and 'faces_noise', figure 4); language and auditory areas during speech[13] (experiments 'speech_basic' and 'speech_lists'); motor regions during hand and finger movements[4,10,14] (experiments 'fingerflex', 'gestures', 'motor_basic', 'imagery_basic', figure 6).

In addition to broadband spectral changes, there are a diverse assortment of cortical oscillations, revealed by low-frequency peaks in the power spectral density (figures 2f, 3).

These oscillations are found throughout human cortex, but rhythms within a named frequency range are not ubiquitous, and, where observed, are idiosyncratic to specific gyral loci. In our ECoG experiments, we found many topologically specific narrow-band oscillations such as a ~10Hz "alpha-rhythm" on the occipital pole (e.g. figure 3b; experiment 'visual_search'), a ~16Hz "beta-rhythm" in peri-central motor cortex (e.g. figure 6e; experiments 'motor_basic', 'fingerflex', 'imagery_feedback', and 'gestures'), and a ~6Hz "theta-rhythm" in the lateral frontal convexity (experiment 'memory_nback'). A few common properties we observed for brain rhythms are: first, the amplitude of oscillations decreases with behavior (e.g. figures 3b,c and 6a); second, these decreases are relatively non-specific and completely overlapping for closely-related behaviors (e.g. different finger movements, figure 6); third, rhythms are phase-coherent over several cm, along gyrally constrained boundaries (e.g. figure 6c,d); and fourth, rhythms entrain broadband spectral changes (an analogue of phase-amplitude coupling, e.g. figure 6e,f)[10,11,13,14]. Interestingly, the alpha-rhythm on the occipital pole is visually apparent in the stimulus-averaged voltage timeseries when images are shown at regular intervals, suggesting that the phase of alpha-oscillations are reset by the onset and offset of visual stimuli (figure 3c; experiment 'visual_search'). Furthermore, this rhythm at stimulus onset appears as a pulsed increase from baseline voltage, rather than a zero-averaged oscillation. This would appear to be evidence for hypotheses of pulsed inhibition by the occipital alpha rhythm for gating by inhibition[15,16]. We should note that narrowband "gamma" oscillations were not generally observed in our series of experiments, and recent ECoG work has indicated that these are an idiosyncrasy of visual stimulus properties rather than a necessary property of cortical processing in the occipital lobe[17].

There are evoked changes in the raw voltage trace (ERPs) associated with behaviors and sensory stimuli in potentials measured from both the scalp and the brain surface[18,19]. We found ERPs (figure 2e) in the ECoG signal associated with frontal motor-cortex electrodes during finger movement (experiment 'fingerflex') and in frontal, occipital, and temporal cortex while viewing visual stimuli[11,12,20] (figures 3&4; experiments 'visual_search', 'memory_nback', 'faces_basic', 'faces_noise'). While ERPs are found in a wide variety of settings, they are sparse on the brain surface and topologically focal, found only in a few non-contiguous brain sites (figure 4).

More sophisticated phenomena observed in the ECoG measurement: A variety of analysis approaches were developed and employed to try to infer underlying structure in the recorded data from this ECoG library and connect it to the statistics of the underlying neuronal population (figure 2), These approaches are included as a code library alongside the data. First among them is a series of fitting techniques that identified and quantified power-law structure in the ECoG power-spectral density (PSD) [2,21](figure 2d, scripts in experiments "fixation_highfreq" and "fixation_pwrlaw"). In order to characterize underlying motifs in the ECoG signal, a singular-value-decomposition based technique was developed for "decoupling" the PSD. An extracted broadband motif from this technique can be applied to a continuous measure of the PSD (the dynamic spectrogram) to track local neuronal activity with high temporal fidelity (figures 3c, 4, and 6g; scripts in "fingerflex" and "visual_search")[1,4]. Complementing these broadband changes, there are oscillations (rhythms) in the ECoG signal. In order to better understand brain oscillations, we developed scripts for quantifying single-trial rhythmic entrainment of neuronal activity ("phase-amplitude coupling"), and calculating single-trial projected coherence between brain sites[11,14] (figure 6; scripts in "fingerflex" and "visual_search"). For comparing and contrasting event-locked raw voltage changes with broadband changes in the spontaneous signal, we developed 'template-projection' techniques to classify behavioural and perceptual states[20] (scripts in "faces_basic") and for designing decoders in control tasks before applying them to a more nuanced, higher-noise, setting[12] (scripts in "faces_noise"). MATLAB scripts in enclosed experimental folders should facilitate extension of these techniques to readers' own data, making use of the library as a control.

Discussion: For new and existing ECoG labs, this library provides a set of basic experiments in a variety of brain areas and behavioral paradigms. To foster discussion about these data, ongoing analyses, and resulting publications, we have setup a Facebook group page ("Kai Miller's Library of electrocorticography data and analyses" - https://www.facebook.com/groups/1741787552810669/).

To create the most stereotyped benchmark possible, as much standardization as possible should be undertaken, utilizing one surgeon, one experimenter, and one amplifier system with the same settings. For this library, just one surgeon (J. Ojemann) implanted all of the electrodes, Surgical technique during implantation can influence the character of the recorded signal. For example, the surgeon might align spaces between electrodes with large vessels and sulci, and electrode position on top of vasculature has been shown to influence signal quality[22]. Presence of blood beneath the array or integrity of the arachnoid membrane likely affects the signal in ways we haven't yet quantified. Similarly, the decision to leave the bone on or sew the grid to the dura (which itself may or may not be tacked to the bone edge) is surgeon specific, and presumably affects the electrode-brain interface, altering the signal.Only one experimenter (KJM) interacted with the patients to record all data in this library. The uncontrolled setting of the hospital room can be countered by the regimented and particular practice of the experimenter, and the patient interaction during task instructions will often contain non-verbal or non-codified elements that the experimenter may not be aware of. For example, the insistence of the experimenter on the way a patient's hand rests during a finger flexion task, taking all of the weight off of the arm and engaging only a few isolated joints may result in a reproducibly more sparse pattern of cortical activation associated with the task. Interactions with the nurses to prevent distraction and interruption can have a palpable effect on data quality. An experimenter will have his or her own particular threshold for aborting an experimental run in the face of distraction or non-participation, and this can greatly affect the end-quality of the data across a cohort of patients. The same set of amplifiers, with the same settings for all experiments, recorded the voltage with reference to a scalp/mastoid electrode, (empirically characterized in the supplement to reference [2]). It was necessary to split the signal between the research and clinical amplifiers, and the mechanism for doing this was the same for all recordings. The same brand of electrodes, with the same sizes and composition was used for all patients. A conjugate approach to the stereotyped benchmark created in this library is to attempt to maximize generalizability by collecting and analyzing data across multiple recording sites, with

multiple surgeons, and multiple experimenters all performing the same set of tasks defined by simple guidelines (one example of which is the ongoing 'Restoring Active Memory' project in development[23]).

Physiology without concrete association to anatomy is difficult to interpret (at best), and so the best effort should always be made to define each electrode position to precise brain structure. For this library, the method for systematic alignment to anatomy was contingent on the type and quality of pre- and post-implantation imaging (figure 5). In some experiments, only co-registration to a standard atlas, using x-rays, was feasible, and scientific claims associated with analysis of these experiments should be limited to general statements about a brain region. For most experiments, it was possible to fuse pre-implant MRI to post-implant CT (with intraoperative photograph verification), allowing concrete association between physiology and specific gyral anatomy. Electrode locations, along with the necessary plotting tools, are available for every patient in every task and allow for straightforward visualization of overlaid analyzed signal properties on brain anatomy.

When assessing data, there are typically multiple ways of approaching a problem, and each will have its' own intrinsic value. For example: Fourier analysis and Fourier-based wavelets are familiar and help us understand the structure of the data in a language we can relate to (figure 3c); conversely, matching pursuit approaches can optimize both how concisely and with what temporal precision the data timeseries is described. The same relationships hold for linear versus non-linear classification approaches. Often techniques may be mapped onto one another, and it can be very useful to examine under what circumstances two complementary methods converge, rather than to embrace one while denouncing the other. With these library-associated analysis scripts, our manuscripts can be reproduced, and methodologies extended to new contexts. As hypotheses emerge in the literature, they can be retested on the library by anyone who wishes. With the existing analysis structure, a backbone is in place, where data are loaded, variables accessed, figures made, and plots cycled through – one may simply insert their code at the appropriate place within scripts and have results plotted on anatomy instantly.

Method

Ethics statement: All patients participated in a purely voluntary manner, after providing informed written consent, under experimental protocols approved by the Institutional Review Board of the University of Washington (#25203). All patient data were anonymized according to IRB protocol, in accordance with HIPAA mandate. The non-anonymized image in figure 1, panel (a), is part of a series taken in anticipation of a newspaper article[24], which the patient provided separate consent for. The

general convention used for anonymization was to pick the initials of US presidents, with predominantly right/left/both sided implantations corresponding to Republican/Democrat/other initials, respectively. No link to the patient's identifying information remain, and, as these are anonymized data, no further information about patients will be made available. All data, cortical renderings, and analysis code are publically available, for use without restriction, at https://searchworks.stanford.edu/view/zk881ps0522.

Patients: All human subjects in the study were epileptic patients at Harborview Hospital in Seattle, WA. Sub-dural grids and strips of platinum electrodes were clinically placed over frontal, parietal, temporal, and occipital cortex for extended clinical monitoring and localization of seizure foci. Placement of electrodes was decided purely based upon clincal requirement, without consideration of possible research involvement. Post-operatively, we examined electrode placement and selected a set of experiments based upon potential for task-responsiveness of covered areas (as anticipated from the fMRI literature). A table in the root directory of the library ("kjm_ECoGLibrary_PatientTaskTable.pdf") describes which patients performed each task.

Recordings: Experiments were performed at the bedside, using Synamps2 amplifiers (Compumedics Neuroscan, Charlotte, NC) in parallel with clinical recording. Stimuli were presented with a monitor at the bedside using the general purpose BCI2000 stimulus and acquisition program[25] (interacting with proprietary Neuroscan software), which also recorded the behavioral parameters (dataglove, joystick, audio, EMG, keystroke, mouse) and synchronized them with cortical voltage measurements. Sub-dural platinum electrode arrays (Ad-Tech, Racine, WI) were arranged as combinations of 8x[4,6,8] rectangular fronto-temporo-parietal arrays and 1x[4,6,8] linear temporal and occipital strips. The electrodes had 4mm diameter (2.3mm exposed), 1 cm inter-electrode distance, and were embedded in silastic. Neither composition nor geometry of these electrodes were changed during the period this library was recorded in. Electrical potentials were sampled at 1000 Hz, with respect to a scalp/mastoid reference and ground (not re-referenced, except in the in the 'vispac' experiment, where it is explicitly stated in the "README" file), with an instrument-imposed bandpass filter from 0.15 to 200 Hz. A preliminary visual examination of voltage tracings was performed to reject channels with obvious artifact or epileptic activity (unless explicitly stated otherwise in the "README" file).

Stimulation mapping: For the "speech_basic" experiment, electrocortical stimulation mapping (ECS)[26,27] was performed to identify language cortex as part of his/her clinical care. In this mapping, 5–10 mA square wave current pulses (1 ms in length) were passed through paired electrodes for up to 3s (less if a response is evoked) to interrupt function, induce sensation, and/or evoke motor responses.

Electrode localization (figure 5)**:** Throughout the course of our ECoG research, our ability to determine the position of each electrode to patient-specific gyral anatomy was variable and depended on the quality of pre- and post-implantation imaging that we were able to obtain in each case. Every attempt was made to perform a systematic alignment to anatomy, in the most accurate way possible. In the library, electrode locations, with the necessary to plotting tools, are available for every patient in every task.

X-ray landmarks for projection of electrode position to an atlas brain rendering (figure 5a): When only post-implant x-ray or CT scout images were available (e.g. when CT and MRI slice thicknesses were too large for volumetric registration), we used a projection technique to estimate the anatomic position of each electrode with respect to a template atlas. For this technique, the line segment drawn from the glabella to the inion (visible on the lateral x-ray) is estimated as parallel to the AC/PC line, with the actual AC/PC line lying at a fixed ratio of transverse distance to the dorsal aspect of the inner skull table[28]. By making the assumption that the inner table of the skull matches the brain surface, the maximum antero-posterior (y) and dorsal-ventral (z) coordinates of the inner table of the skull can be normalized to a template brain volume, with identification of a unique medio-lateral (x) coordinate for each y,z along a curved surface that approximates the inner aspect of the dura (a convex hull generated from the cortical surface of the template brain). All electrode positions can be subsequently plotted on the atlas brain to estimate their relationship to underlying anatomy. A

MATLAB-based software package (the "LOC package") was constructed to automate this process, and allows for easy plotting of data to the atlas brain[6] (folder 'loc' in the root directory of the library).

Projection of post-implantation CT to fully rendered individual patient brain rendering from volumetric MRI (figure 5b): A thin-cut post-operative CT head is fused to a pre-operative structural MRI, and the electrode positions are determined in the coordinate space of the CT. A cortical surface mesh is generated from this pre-operative MRI, using either the freesurfer[29] or spm[30,31] environment. A convex hull generated from the cortical surface, and electrodes (often lying below the surface) are projected to this convex hull so that they sit at the cortical surface. This fusion is performed using the CTMR package[7] (folder 'ctmr' in the root directory of the library), which has been demonstrated to accurately localize the electrode positions within an error of ~4mm (the size of the clinical ECoG electrodes).

Reslicing volumetric CT to thick cut MRI (figure 5c): In some instances, we had non-volumetric, thickly cut MRI axial sections, but high-resolution, volumetric, post-implantation CT scans. In this case, the CT was then interpolated and resliced into the T1 axial MRI. Electrodes were then identified in this mutual space on each axial slice so that their positions were known with respect to gyral anatomy from MRI[5]. A set of simple MATLAB scripts for generating co-registrations and navigating the axial co-registration are in the folder 'xs_files' in the root directory of the library.

X-ray landmarks for projection of electrode position to fully rendered individual brain rendering from volumetric MRI (figure 5d): When high-resolution MRI is available, but only with post-implant x-ray or CT scout images. A 3-D cortical mesh is obtained using either the freesurfer[29] or spm5[30,31] environment. Using a layer-based image processing program (photoshop or other), the mesh is rotated into a pure lateral or anterior-posterior view, and then the x-ray is manually rotated and scaled so that the brain outline fits the inner table of the skull. By identifying electrodes on a separate layer from the x-ray, while making the brain rendering images selectively transparent, electrode positions can be directly visualized at the appropriate points on the brain rendering[32].

Tasks (Table 1): A variety of tasks were performed with each patient we were able to work with, tailored to the approximate functional correlation expected from the functional imaging literature. Table 1 summarizes these tasks, with pointers to the appropriate folders in the data library. Within each of these folders is a file named "README_[folderName]_dataset_notes.pdf" that comprehensively describes:

- the task in detail, with the subfolder locations and naming conventions of data files
- specific descriptions of each variable in the corresponding datasets,
- instructions on the corresponding behavioral recordings (dataglove, joystick, audio, EMG, keystroke, mouse),
- stimulation results (when applicable), along with file subfolder locations,
- electrode location and brain rendering file subfolder locations,

For the tasks with manuscripts published from analyses of the data, the relevant manuscript(s) are included in a "refs" subfolder.

Signal Processing illustrated in this manuscript (Note that there are other methods employed in the analyses of many dataset folders, and these are described in the reference manuscripts contained therein). The corresponding scripts for the finger flexion implementation of these methods are contained within the 'fingerflex' folder:

Pre-processing: To reduce common artifact, the raw potential, $V_n^0(t)$, measured at each electrode n was re-referenced with respect to the common average of all N electrodes (script 'car'), $V_n(t) = V_n^0(t) - \frac{1}{N}\sum_{i=1}^{N} V_i^0(t)$, (channel label, n, henceforth dropped). Electrodes with significant artifact or epileptiform activity were visually rejected prior to common averaging (as has been done for all datasets in the library) and omitted from further analysis.

Power spectral snapshots (script calc_dg_spectra.m in subfolder 'dc_files'): A set of epochs surrounding events τ_q (image presentations or finger movements) were extracted from $V(t)$; each epoch was of duration T =1s, $(\tau_q - T/2) < t < (\tau_q + T/2)$. The epochs were sorted according to cue type q, and labeled by their event markers τ_q. The power spectral density (PSD) of the epoch flanking time τ_q was calculated as:

$$P(f,q) = \left| \frac{1}{\sqrt{T}} \sum_{t=-T/2}^{+T/2} V(\tau_q + t) H(t) e^{t 2\pi i f t} \right|^2$$ with Hann windowing function[33], $H(t) = \frac{1}{2}\left(1 + \cos\left(\frac{2\pi t}{T}\right)\right)$.

Dynamic power spectral measures (script dg_tf_pwr_rm.m in subfolder 'dc_files'): Time-frequency approximations (dynamic spectra) were made using both a wavelet and a Hilbert transform approach (although these can be equivalent under specific conditions, they are used in a way such that they fill different roles and reflect very different quantities in this study).

Wavelet approach: A Morlet wavelet[34] of the form: $\psi(\tau,t) = \exp\frac{i2\pi t}{\tau} \exp\frac{-t^2}{2\tau^2}$ was multiplied with a sliding window of the voltage timeseries to get a time-frequency estimate for every $f = 1/\tau$:

$$\tilde{V}(1/\tau, t) = \sum_{t'=-5\tau/2}^{5\tau/2} V(t+t')\psi(t',\tau)$$

A total of 5 cycles (5τ) were used to estimate the amplitude and phase of the signal at each frequency for every point in time. In this way a time-varying Fourier component $\tilde{V}(f,t) = r(f,t)e^{i\varphi(f,t)}$ is obtained at each Hz, with fixed uncertainty between the confidence in the estimate of the instantaneous amplitude and phase versus the confidence in temporal resolution.

Event-locked power spectral density: This time-frequency approximation can be used to calculate mean power in relation to the onset of events from each epoch:

$$\overline{P}^{v}(f,t_w) = \frac{1}{N_v} \sum_{\tau_v} \left(\frac{\left|\tilde{V}(f, t_w + \tau_v)\right|^2}{\frac{1}{N_{t'}} \sum_{t'} \left|\tilde{V}(f,t')\right|^2} \right)$$

Where τ_v denote onset times of image presentations or the first finger movement in response to each cue (total N_v), and t' denote inter-stimulus times (total $N_{t'}$). The peri-image presentation or peri-movement time window of interest is denoted t_w (figure 3c).

Hilbert transform approach (script get_rhythm_dist.m): A complex signal to reflect the timecourse of a functionally-relevant frequency-range band was constructed as follows: The signal $V(t)$ was band-passed using a 3rd order Butterworth filter for a specific range, to obtain the "band-limited" potential, $V(F,t)$, where F denotes the frequency range (e.g. for "beta", $F \rightarrow$ [12-20Hz], etc.). A complex, analytic, signal, $\tilde{V}(F,t) = V(F,t) + iV^{IM}(F,t)$ was constructed using the Hilbert transform (e.g. such that the new complex signal satisfies the Cauchy-Riemann conditions for analyticity at all times). This signal may also be expressed in polar notation: $\tilde{V}(F,t) = r(F,t)e^{i\phi(F,t)}$. The "analytic amplitude" of the frequency range F at time t is $r(F,t)$ and the "phase" is $\phi(F,t)$. The interpretation of ϕ is intuitively difficult, but the most concrete understanding in our context is that the rhythm captured by range F is most surface-positive at $\phi = 0$, and most surface-negative at $\phi = \pi$ or $-\pi$ (this interpretation would not be valid for bipolar re-referenced data). Note that ϕ becomes poorly defined as $r \rightarrow 0$.

Decoupling the cortical spectrum to separate rhythmic activity away from broadband change: The decoupling process is described and illustrated in full detail in the main text and supplement to the Miller, et. al. 2009 manuscript[4].

Principal component decomposition of spectral change (script call_subs_processing.m, annotated step 5): The samples of the PSD, $P(f,q)$, (total N_q), were first normalized prior to decomposition.

$$\hat{P}(f,q) = \ln\left(P(f,q)\right) - \ln\left(\frac{1}{N_q}\sum_q P(f,q)\right)$$

Then, a singular value decomposition is used to determine the eigenvalues λ_k and eigenvectors $\vec{e}_k$ of the correlation matrix $C(f,f') = \sum_q \hat{P}(f,q)\hat{P}(f',q)$.

These eigenvectors, $\hat{C}\,\vec{e}_k = \lambda_k\,\vec{e}_k$, the ``Principal Spectral Components" (PSCs), reveal which frequencies vary in power together, and are ordered by magnitude of corresponding eigenvalue: $\lambda_1 > \lambda_2 > \cdots > \lambda_{N_f}$ ($N_f \equiv$ number of frequencies and also eigenspectra). If we define the rotation matrix $A(f,k) = (\vec{e}_1, \vec{e}_2, \cdots, \vec{e}_{N_f})$, then the projection, $W(k,q)$, of each individual original spectrum in the ensemble onto the new basis vector $\vec{e}_k$ is $W(k,q) = \sum_f A(k,f)\hat{P}(f,q)$.

The inverse rotation matrix $\hat{A}^{-1}$, $\hat{A}^{-1}\hat{A} = \hat{I}$, allows us to compare and visualize specific subsets of PSC components with the original full spectrum in frequency space. The 2nd to 4th PSCs typically capture rhythmic power spectral phenomena, and power spectra can be reconstructed with and without this rhythmic influence: $\hat{P}_\Xi(f,q) = \sum_{k\in\Xi} A^{-1}(f,k)W(k,q)$.

If the 2nd to 4th PSCs are omitted, $\Xi \rightarrow [1,5 \ldots N_f]$, then PSDs can be reconstructed where changes in rhythmic spectral phenomena are mostly removed (although there may be residual variance in the decomposition, or some rhythmic influence in all cases). If $\Xi \rightarrow [2\text{-}4]$, then PSDs can be reconstructed where changes in rhythmic spectral phenomena are mostly isolated.

The timecourse of broadband spectral change (script gen_pc_all.m in subfolder 'dc_files'): The time-dependent, normalized, dynamic power spectrum, $\hat{P}(f,q)$, can be obtained in parallel fashion to the spectral snapshots.

$$P(f,t) = \frac{\left|\tilde{V}(f,t)\right|^2}{\frac{1}{N_t}\sum_t \left|\tilde{V}(f,t)\right|^2}, \text{ and } \hat{P}(f,t) = \ln\left(P(f,t)\right) - \ln\left(\frac{1}{N_t}\sum_t P(f,t)\right).$$

The reflection of the 1st PSC ($\vec{e}_1$) in the dynamic spectrum can be estimated by projecting the dynamic spectrum onto it: $\ln A(t) = \sum_f e_1(f)\hat{P}(f,t)$. We call it $\ln A(t)$ here, because we believe it approximates the logarithm of the timecourse of the coefficient of a power law in the cortical spectrum[2] of the form $P(f,t) = A(t)f^{-\chi}$; it is smoothed with a Gaussian window of 50ms standard deviation, z-scored, and exponentiated to obtain the "broadband" traces. Because the quantity $\ln A(t)$ is approximately normal-distributed, we express it in Z-score units, and, for notational brevity, denote it $\chi(t)$ in connection to the broadband power law it reflects. Note that the smoothing and re-exponentiation apply only to the broadband illustrations, and not to calculation of coupling to rhythmic phenomena (figure 3c).

Broadband coupling to low-frequency phase*:* The quantification of rhythmic entrainment, phase coherence, and spatial overlap, are described and illustrated in full detail in the main text and supplement to the Miller, et. al. 2012 manuscript[14].

Phase-coupling Palette (script gen_pac_matrix_all.m): The coupling between rhythm phase and local cortical activity is first illustrated by calculating the average log-broadband amplitude $\chi(t)$

as a function of the wavelet-obtained Fourier phase, $\varphi(f,t)$, at each Hz. When used to build up a "palette" displaying many frequencies at once, a full picture of rhythmic entrainment emerges (figure 6e).

The coupling vector and trial-by-trial statistics (script get_rhythm_dist.m): Because the underlying physiology may have slight cycle-to-cycle variability and non-sinusoidal shape, there may smearing across frequencies in this coupling[14]. A rhythm spanning multiple frequencies, F, on the phase-coupling palette can be further characterized by applying a band-pass filter and Hilbert transform to $V(t)$ to obtain the time-dependent Hilbert phase, $\phi(F,t)$. The average log-broadband amplitude $\chi(t)$ as a function of $\phi(F,t)$ in small phase bins, ϕ_k, is denoted χ_k. This allows for the calculation of a "coupling vector" by taking the dot product $Z_{\mathrm{mod}}e^{i\phi_c} = \frac{1}{2K}\sum_k \chi_k e^{i\phi_k}$. Z_{mod} is the magnitude of coupling between phase of the rhythm, F, and the log-broadband amplitude (because we z-score $\chi(t)$, Z_{mod} is roughly the amount of variation in the z-score that can be explained for by modulation by the phase of the rhythm concerned), and ϕ_c is the preferred phase of this interaction.

This can be calculated on a trial-by-trial basis, breaking up the data into smaller epochs, n, of continuous movement or rest and calculating a coupling vector for each epoch. Because these individual-epoch $Z_{\mathrm{mod}}(n)$ must be non-negative, the distribution must be significantly greater than zero even when there is no underlying coupling of consistent phase. A true estimate of distributions of coupling should perform a correction that assumes the phase of correction should be reliable.

This is done by projecting each trial's coupling vector onto the preferred phase of the global average coupling vector in the following fashion: First, the global average coupling vector is calculated $\widehat{Z_{\mathrm{mod}}}e^{i\widehat{\phi_c}} = \frac{1}{N}\sum_n Z_{\mathrm{mod}}(n)e^{i\phi_c(n)}$. Note that $\widehat{Z_{\mathrm{mod}}} \neq \overline{Z_{\mathrm{mod}}(n)}$ nor $\widehat{\phi_c} \neq \overline{\phi_c(n)}$. Second, the individual epoch coupling vectors are projected onto the phase of the average coupling vector to obtain corrected modulation values, $Z^c_{\mathrm{mod}}(n) = Z_{\mathrm{mod}}(n) * \cos\left(\phi_c(n) - \widehat{\phi_c}\right)$. The quantities $Z^c_{\mathrm{mod}}(n)$ can be negative or positive and can therefore have a distribution significantly overlapping with zero (indicating an absence of reliable coupling). Furthermore, distributions of different types of epochs (visual stimuli, finger movement, rest, etc), each with their own preferred average phase of coupling $\widehat{\phi_c}$, can be compared versus one another or versus zero (as in figure 6f).

Phase coherence of rhythms (script task_cohere_corr.m): The pair-wise phase coherence of an electrode array relative to an internal reference "seed" site was examined. The complex portion was retained because, in the setting of a dominant, distributed, rhythm, the choice of common average reference introduces phase coherence with brain regions where the rhythm is otherwise absent. However, introduced phase coherence will be approximately π radians out of phase with the dominant rhythm. The method to measure phase coherence employed in this paper reflects only a part of the full phase coherence but is sufficient to address the spatial distribution of pairwise phase coherence. For the complex rhythm $\tilde{V}_a(F,t)$ ("seed" electrode a, frequency range $F \rightarrow$ [12-20Hz]), the complex phase coherence with another electrode b, for epoch q (during time t_q) is: $Q_{ab}(F,q) = \frac{1}{T_q}\sum_{t_q} e^{i\left(\phi_a(F,t_q) - \phi_b(F,t_q)\right)}$. In figure 6c, the "seed" electrode is the motor electrode with the strongest index finger movement-associated broadband change. In order to illustrate the boundaries and dominant motifs in this phase coherence more clearly, the complex phase coherences for all electrodes in an array are projected onto a unit vector in the direction of the phase lag of the site that has the highest absolute phase coherence: if we denote the pair-wise complex phase coherence $Q_{ab}(F) = r(a,b)e^{i\phi(a,b)}$, and the phase shift of the pair with maximum phase coherence as $\phi_{\max}$, then the

projected phase coherence is $Q_{ab}^{c}(F) = r(a,b)\cos(\phi(a,b) - \phi_{\max})$, (figure 6c, shipwheel symbol denotes seed site).

Quantifying overlap in spatial distribution (script coh_overlap_work.m): In order to quantify the overlap between spatial extent and degree of change in two different measures, we use a resampling metric. For measures of types X_n and Y_n (n denoting electrode), the simple dot product between the two gives a "true" overlap, $O_{XY}^{T} = \sum_{n} X_n Y_n$, that can be compared to surrogate distributions for quantification of magnitude and significance. The spatial overlap metric is the true overlap divided by the maximum possible overlap if electrode labels were reassigned: $Overlap = \frac{O_{XY}^{T}}{\sum_{n} X'_n Y'_n}$, where X'_n denotes the distribution X_n, re-sorted in ascending order. To estimate the significance of this overlap using resampling, the electrode labels are then randomly scrambled ($n \xrightarrow[k]{} m$), and a surrogate overlap is obtained $O_{XY}^{S}(k) = \sum_{m} X_m Y_m$. This is done 10^6 times. A p-value for the significance of the overlap is the percentage of O_{XY}^{S} that are greater than O_{XY}^{T} (or the percentage that are less in the case that $O_{XY}^{T} < 0$, figure 6b&d).

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Figure captions:

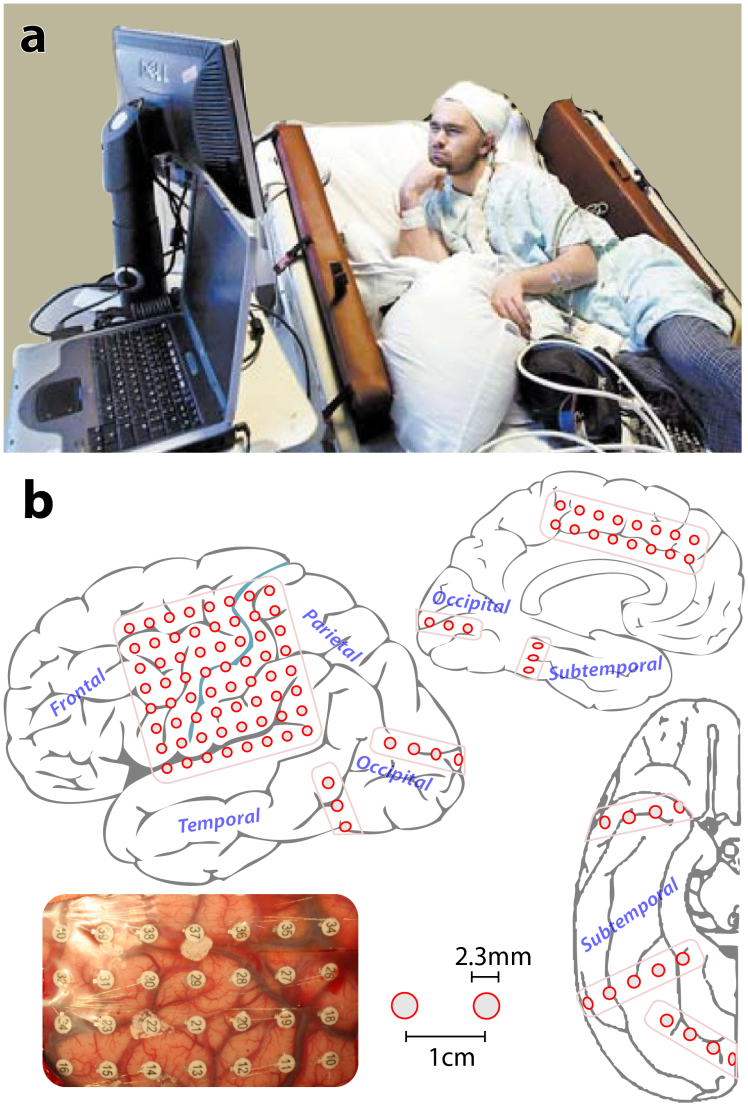


Figure 1: Experimental recordings. (a) Experiments were performed at the bedside, splitting the ECoG signal to separate clinical and research amplifiers. Patients performed behaviors based upon cues from a bedside monitor. **(b)** Platinum electrocorticography (ECoG) electrode arrays were configured as linear strips or grids, and embedded in silastic with 1cm spacing (4mm diameter with 2.3mm exposed). They were placed along the subdural space via open craniotomies. The non-anonymized image in panel (a) is associated with a newspaper article[24].

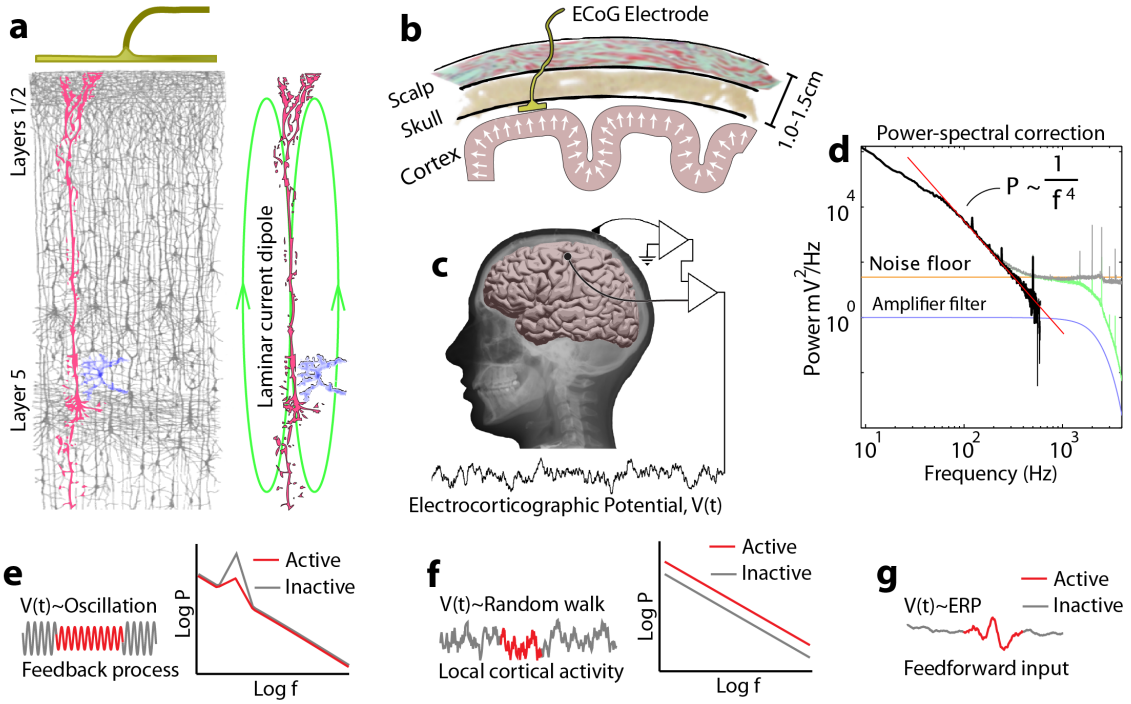


Figure 2: The ECoG signal. (a) The ~5mm^2 of cortical surface beneath each ECoG electrode contains about half a million neurons[35], each of which contributes a current dipole produced by charge movement across the cellular membrane[36]. **(b)** ECoG electrodes are surgically placed subdurally on the cortical surface and are later common average re-referenced after epileptic and artifactual channels are rejected. **(c)** ECoG potentials are initially referenced against an external scalp or mastoid reference. **(d)** After re-referencing, the ECoG potential power spectral density (PSD) can be quantified (green), and then corrected for both the built-in amplifier filter (blue, intermediate PSD in gray) and intrinsic noise floor (orange, final PSD in black). This final PSD is shown to have a power-law shape, tending to $P \sim 1/f^4$ at high frequencies (fit in red, data and analyses from experiment 'fixation_highfreq'). **(e)** Synchronized neural activity in response to feedback loops can be observed in narrowband oscillations, which are peaked in the power spectral density. **(f)** Aggregate neural activity from a region is revealed by behaviorally-associated changes in 1/f shape of the PSD[2]. **(g)** Synchronized neuronal activity in response to a timed stimulus can be observed in the event-averaged raw potential (ERP) and is presumed to be due to feedforward input from a different cortical region. Panel (d) reproduced from reference [2].

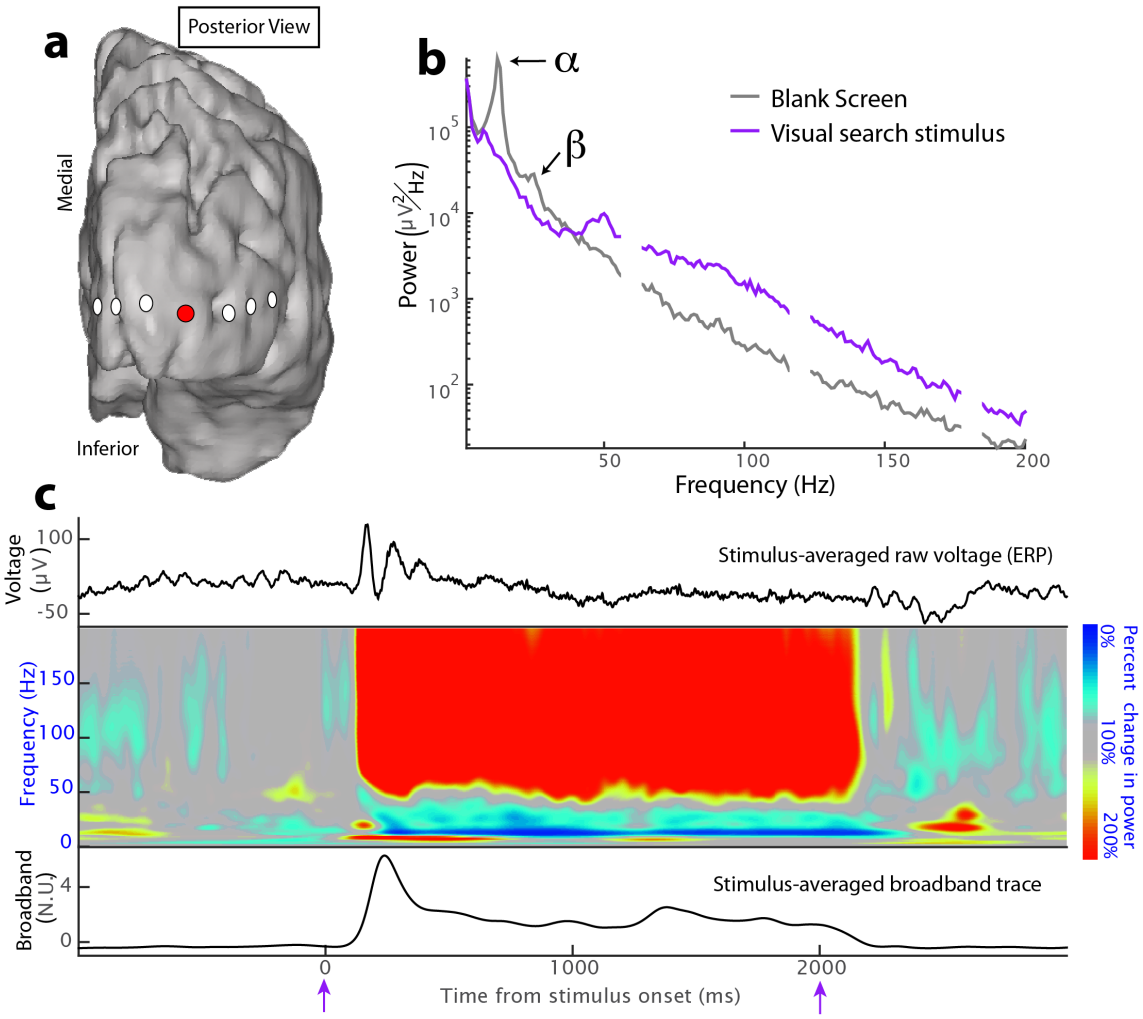


Figure 3: An example of basic ECoG changes when a cortical area becomes active. The patient engaged in a simple visual search task for 2s blocks, interleaved with 2s blank screen interstimulus periods (Experiment 'visual_search'). **(a)** Physiology from an occipital pole electrode (red) is examined in (b) and (c). **(b)** Power spectral density during visual search (purple), is compared with blank screen (gray). There are prominent oscillations in the alpha and beta range that decohere during visual search. **(c)** Event-averaged changes, from 1s prior to each visual search to 1s after (purple arrows on bottom indicate onset and offset of visual search task). The top trace is the event-locked average raw potential (ERP). The fact that there are visually apparent alpha oscillations in this trace during blank screen periods reveals that the stimuli 'reset' the alpha oscillation (e.g. there is a fixed ISI and it does not average away, so it's phase must be task locked). The middle plot is an event-locked averaged spectrogram, showing power decrease in the alpha and beta range, and broadband increases above that. The bottom broadband trace reveals a sustained event-locked broadband increase throughout the visual search period.

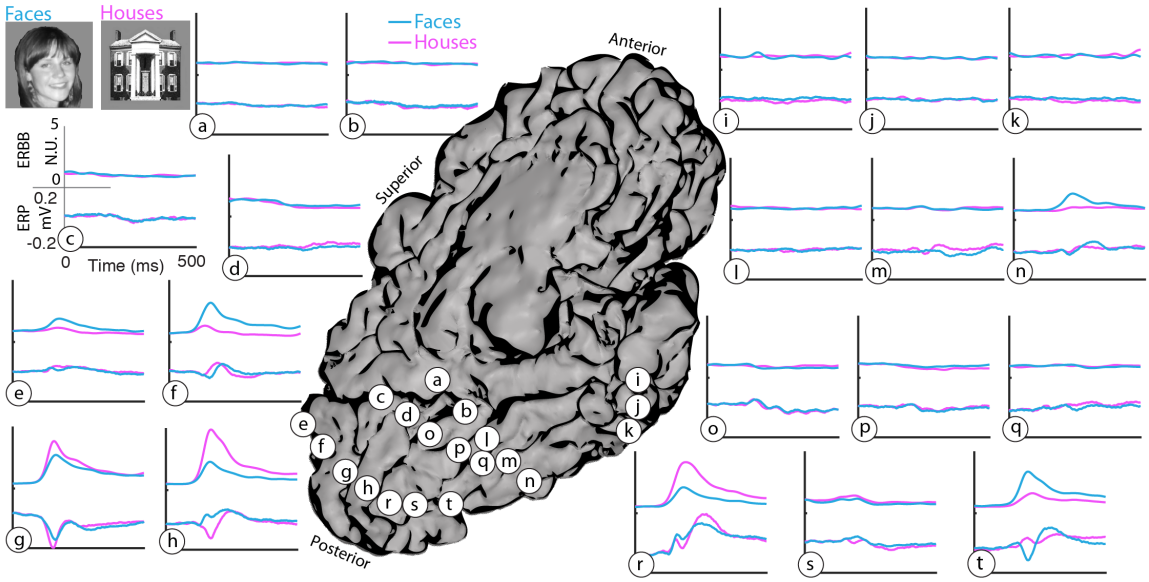


Figure 4: A comparison of broadband and ERP changes across the ventral brain surface, during a task where simple face and house pictures (such as those in top left) are shown for 400ms at a time, interleaved with 400ms blank screen periods. As noted in the panel for site (c), upper traces show broadband responses, and lower traces show ERPs. ECoG ERPs are highly focal. Furthermore, face-selective ERPs are highly polymorphic, and there is no simple mapping from broadband selectivity to ERP selectivity[20] (Experiment 'faces_basic').

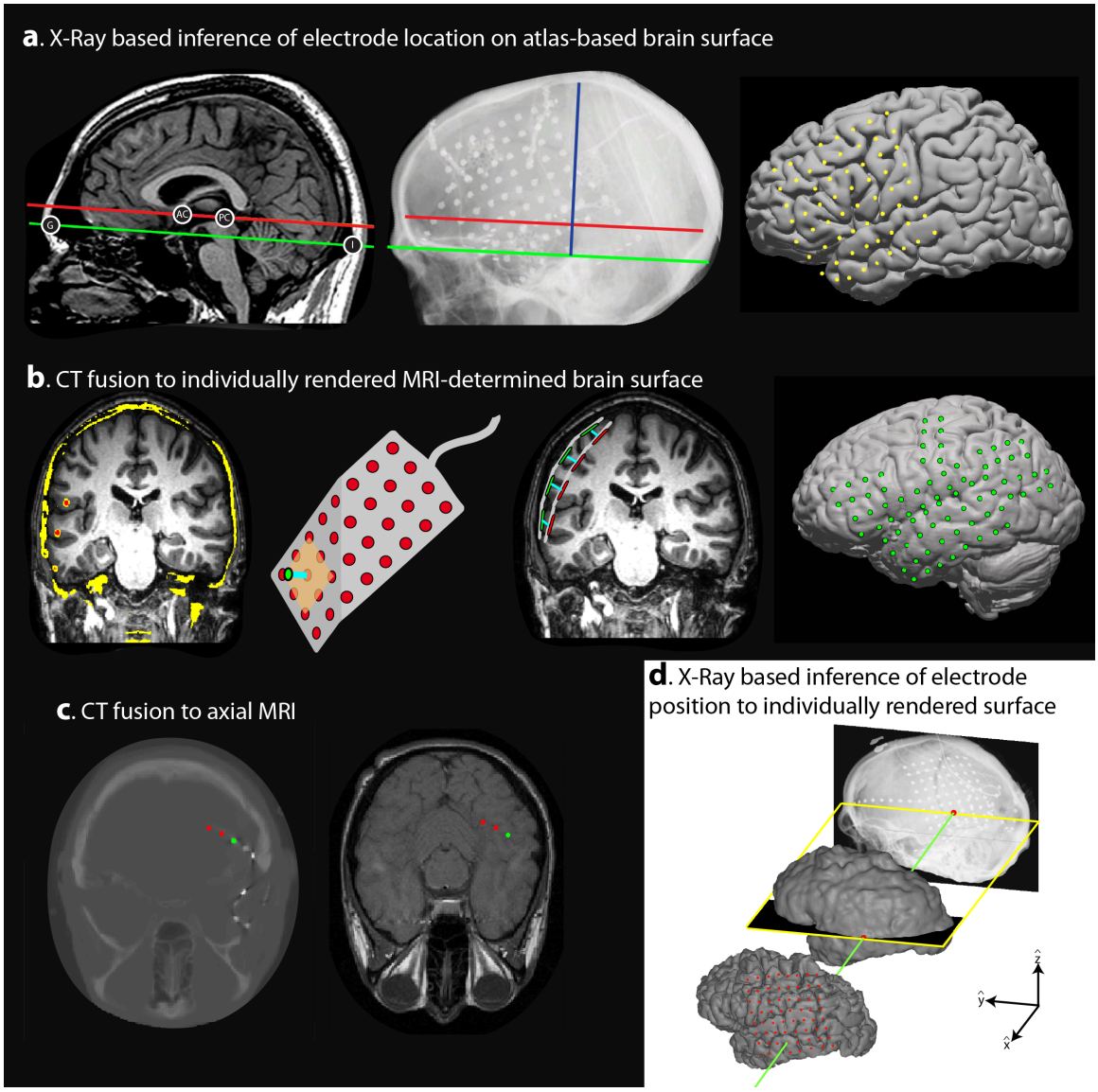


Figure 5: Methods for localizing electrode position to brain anatomy. (a) Landmarks on X-ray can be used for localization to an atlas brain rendering (Folder 'loc'). **(b)** Patient's brain surface can be rendered from MRI and fused to post-implantation CT (Folder 'ctmr'). **(c)** Electrode locations on CT can be matched to MRI cross-section (Folder 'xs_tools'). **(d)** Plain film X-ray can be used to estimate electrode position on MRI-based brain surface rendering. Panels reproduced from references [5-7,32].

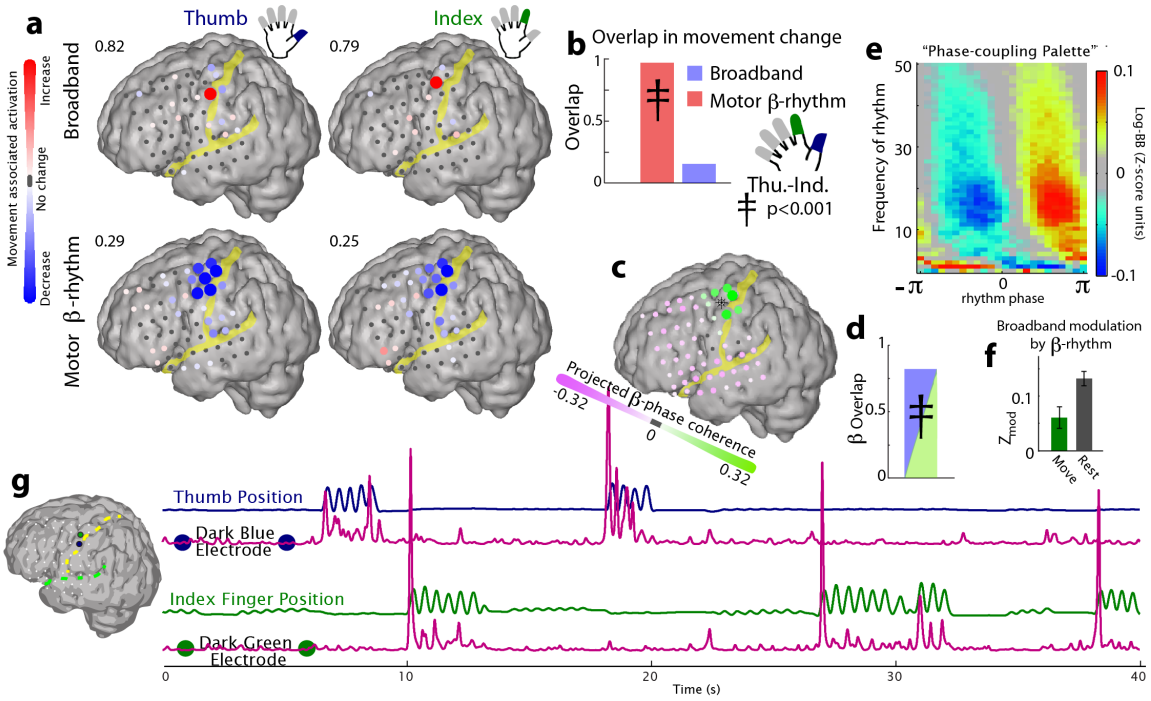


Figure 6: Illustrating a range of analyses for a specific setting - A close look into movement-related ECoG changes during a finger movement task. All of the MATLAB scripts and data to perform these analyses are in the experiment "fingerflex". **(a)** Changes in broadband and motor β-rhythm (12–20Hz power) at different cortical sites for movement of the thumb and index finger. Colors denote a signed r^2 measurement of increases and decreases in power with movement relative to rest (individually scaled with maximum noted to upper left of each plot). **(b)** Quantification of spatial overlap between changes associated with finger-movements. (1 is maximum possible overlap and 0 is no overlap, with significance determined by resampling, 10^5 iterations, $p<0.001$ that the overlap happened by chance is double cross). **(c)** The pairwise complex phase coherence in 12-20Hz beta oscillation signal is calculated between the site of greatest broadband change during index finger movement (top right in (a), designated by a shipwheel symbol), and every other electrode. This complex coherence was then projected onto the phase of the site of maximum absolute phase coherence to obtain the map shown. **(d)** Quantification of spatial overlap between projected beta oscillation phase coherence (in (c)) and index finger movement associated beta power change (bottom right in (a)), significant by resampling to $p<0.001$. **(e)** log broadband amplitude plotted as a function of frequency and phase ("Phase-coupling Palette") for the electrode with the most significant broadband change during thumb movement. **(f)** Coupling vector amplitudes for thumb movement (green) and rest (gray) periods. **(g)** Traces of thumb (dark blue) and index (green), with corresponding timecourse of broadband spectral change (pink; approximated by the projection of the 1st PSC to the wavelet-obtained dynamic spectrum) for 2 cortical sites. Figure partially reproduced from[14].

Experiment Name	Description	Anatomy	N	Ref(s)
fixation_pwrlaw	The **basic baseline fixation task** was performed by the patients as they stared for 2 to 3 minutes at an "X", on the wall 3 m away. Data were examined for power-law structure.	Fronto-parietal-temporal grids	20	2
motor_basic	Patients performed simple, repetitive, **movements of the hand** (synchronous flexion and extension of all fingers) **or tongue** (opening of mouth with protrusion and retraction of the tongue, i.e., sticking the tongue in and out), at ~1-2 Hz.	Fronto-parietal grids	19	10
fingerflex	Patients were cued with a word displayed on a bedside monitor to repetitively **move individual fingers** (contralateral to electrode array) during 2s cue periods while finger position was recorded with a dataglove.	Fronto-parietal grids	9	4,14
joystick_track	The patient used a **joystick to track** a counter-clockwise moving target (hand was contralateral to electrode array).	Fronto-parietal grids	4	37
gestures	A series of hand motor tasks were performed where patients performed **cued simple individual finger movements, cued gestures, or self-generated gestures**, while finger position was recorded with a dataglove (hand used was contralateral to grid).	Fronto-parietal grids	5	Data not published
imagery_basic	Patients first performed an overt hand and tongue cued movement task. They subsequently performed a task **imagining making identical movements**.	Fronto-parietal grids	7	38
imagery_feedback	Patients first performed a cue-based movement task, then a **cue-based imagery** task, then a **one-dimensional cursor control feedback** task based upon the imagery.	Fronto-parietal grids	4	38
speech_basic	Patients participated in a **simple verb-generation task**, where nouns were presented on a screen, and the patient would either read the noun or speak a verb that was connected to the noun: for example, if the cue read "ball", the patient might say "kick".	Fronto-parietal-temporal grids	7	13
memory_nback	After a period of baseline fixation, patients performed an n-back **working memory task** using sequences of house pictures. First they identified when a 'target' house picture appeared (0 back). Second, they identified when a picture had been shown twice in a row (1 back). Finally, they identified when a picture had been repeated with a distracting image in between (2-back).	Right lateral frontal grids	4	Data not published
visual_search	In the **visual search task**, pictures were shown of arrays of colored squares. The patient would visually navigate between the squares depending on a cue, and report the color of a targeted square.	Occipital strips, medial & lateral	5	11
faces_basic	**Simple face/house pictures** were shown in random order for 400 ms each, with 400ms inter-stimulus blank screen, and patients reported an upside-down house.	Subtemporal strips	14	5,20
faces_noise	After performing a simple face and house picture viewing task, patients then performed a face-detection task using **phase-scrambled pictures of faces and houses**.	Subtemporal strips	7	12
fixation_PAC	The basic **baseline fixation** task (as in fixation_pwrlaw) was examined for broadband entrainment on oscillations.	Fronto-parietal-temporal grids	10	14
fixation_highfreq	The **basic baseline fixation task** (as in fixation_pwrlaw) was performed while **recording at 10kHz**. Data were examined for power-law structure.	Fronto-parietal-temporal grids	4	2
mouse_track	Patients used a **mouse** on a mousepad to track a **counter-clockwise moving target** (hand contralateral to grid).	Fronto-parietal grids	4	Data not published
speech_lists	These files form a set of **speech data for lists of nouns** that were first read directly, and then used to produce **associated action verbs**. There were 2 noun lists, and each list was presented 3 times for noun reading, then 3 times for verb production.	Fronto-parietal-temporal grids	4	Data not published

Table 1: List of experiments, with identically named library folders downloadable at https://searchworks.stanford.edu/view/zk881ps0522. References in rightmost column "Ref(s)" correspond to the reference in which the relevant experimental data were originally described. The "N" column denotes the number of patients in each experiment.